DevSync

Secure AI Project Collaborator

Literature Review

Project Type	Graduation Project (Flutter mobile application)
Team	6 members, all Flutter developers
Project Manager	Mohraeel (Project Manager and Flutter Lead)
Platform	Android (Android App Bundle), Flutter + Firebase
Document Version	1.1 - October 2026

Contents: 1. Feedback and Evaluation | 2. Suggested Improvements | 3. Final Grading Criteria | 4. Supporting Literature Review (Related Work)

1. Feedback and Evaluation

1.1 Lecturer's Assessment (to be completed)

This section is left for the lecturer's assessment of the project.

Area	Lecturer Feedback	Date
Proposal and planning		
Literature review		
Requirements		
System analysis and design		
Implementation		

1.2 Feedback Tracker

The team records every piece of feedback (from the lecturer or mentors), the action taken, and its status.

Date	Source	Feedback	Action Taken	Status

2. Suggested Improvements

2.1 Improvements Identified by the Team

- Deeper repository analysis (commit content, code review activity) for more accurate skill estimation.
- Collaborative filtering once enough interaction data has been collected, combined with the current content-based approach.
- iOS release and a web version of the platform.
- Real payment handling for the Pro plan, and Pro-only features such as advanced match analytics.
- A richer community area (events, discussion, mentoring) and a defined Digital Twin of the developer's skills.
- Integration with CI and issue trackers so tasks stay in sync with the real repository.
- Recommendation feedback with reasons (thumbs up or down and a "why not" question) and adaptive matching that learns from accepted and rejected recommendations (features 37, 38, 39).
- An AI learning path with a week-by-week roadmap generated from a user's skill gap (15), and an AI project assistant chatbot that finds projects and explains matches (56).
- A developer growth dashboard that shows skill progress across projects (46).
- Project bottleneck detection that finds the roles or tasks holding a team back (42).
- A GitHub activity score based on verifiable activity, presented as evidence and not as a judgment of a person's trustworthiness (29).
- An AI team builder and a find-missing-person search that suggest the best team or the best candidates for an open role (21, 51).

2.2 Improvements Suggested by the Lecturer (to be completed)

Suggested Improvement	Priority	Status

Suggested Improvement	Priority	Status

3. Final Grading Criteria

Marks are not yet confirmed. Fill in the breakdown when the lecturer provides it.

3.1 Grading Breakdown (to be confirmed with the lecturer)

Component	Marks	Notes
Documentation	TBD	Planning, literature review, requirements, technical documentation and README
Implementation	TBD	Working features, code quality, Git workflow
Testing	TBD	Test plan, results and bug handling
Presentation	TBD	Slides, live demo and Q&A
Total	TBD	

3.2 Official Submission Deadlines (DEPI)

Item	Deadline
Project Planning and Management	16 Oct 2026
Literature Review	16 Oct 2026
Requirements Gathering	16 Oct 2026
System Analysis and Design	6 Nov 2026
Implementation (Source Code and Execution)	30 Nov 2026
Final Presentation, Testing and Reports	4 Dec 2026

All documents are uploaded to the team GitHub repository.

4. Supporting Literature Review (Related Work)

4.1 Purpose

This review looks at existing platforms and research related to DevSync: finding open-source projects and collaborators, recommending them with AI, and forming balanced teams. Its goal is to show what already exists, what is missing, and where DevSync fits.

4.2 Related Platforms

The descriptions below are based on general public knowledge of each platform. The team should re-check current features before submission.

Platform	What It Offers	Limitation Compared to DevSync
GitHub Explore, topics and "good first issue" labels	Browsing repositories by topic and finding beginner-friendly issues; Discussions for communication	Search is manual; no personalized matching to a user's skills; no explanation of fit; no team-role awareness
Up For Grabs and CodeTriage	Curated lists and subscriptions of beginner-friendly issues, filterable by language or tag	Issue-level only; does not analyze the user's GitHub activity; no team formation or collaboration tools
Devpost (teammate search)	Finding teammates for hackathons and showcasing projects	Tied to events; manual browsing; not designed for continuing open-source collaboration
Co-founder matching platforms (for example Y Combinator Co-Founder Matching)	Profile-based matching between people building startups	Targets startups, not open-source; no technical skill-gap analysis from repository data
LinkedIn and Behance	Professional networking and portfolio presentation	No repository-level technical signals; no project matching or task collaboration

4.3 Research Background

- **Newcomer barriers in open source.** Steinmacher et al. (2015) reviewed the barriers newcomers face when joining open-source projects, including difficulty finding a suitable first task and getting support. DevSync addresses this by recommending projects that match the user's level and by showing skill gaps.
- **Recommending beginner-friendly tasks.** Xiao et al. (2022) studied automated recommendation of good first issues on GitHub, showing that data from repositories can be used to guide newcomers. DevSync extends the idea from issues to projects and roles.
- **Team formation.** Lappas, Liu and Terzi (2009) formalized the problem of selecting a team of experts that covers a set of required skills. This supports DevSync's missing-role detection and role generation.
- **Explainable recommendation.** Zhang and Chen (2020) survey explainable recommender systems and discuss how explanations improve transparency and user trust. This is the basis of DevSync's Explain AI feature.
- **Prompt engineering.** Research on large language model prompting, such as Wei et al. (2022), shows that prompt structure strongly affects output quality. DevSync therefore treats prompts as testable artifacts, with prompt evaluation and optimization based on user feedback.
- **Mobile security.** The Android software stack uses application sandboxing and a runtime permission model. DevSync follows these principles for permission handling, data storage and release hardening (code obfuscation).

4.4 Gap Analysis

Need	Existing Solutions	How DevSync Closes the Gap
Personalized project matching	Manual search and generic lists	AI matching based on GitHub activity, level and declared skills

Need	Existing Solutions	How DevSync Closes the Gap
Trust in recommendations	Rankings with no reasoning	Explain AI shows why a project matches
Knowing what to learn	Not offered	Skill gap and cross-skill match for each project
Balanced teams	Left to the project owner	AI role generator and missing role alerts
From match to collaboration	Users move to other tools	Requests, invites, chat, workspace and tasks inside the app
Continuous improvement	Static algorithms	Prompt evaluation and optimization from user feedback
Secure mobile delivery	Mostly web-based tools	Android app with runtime permissions, obfuscation and Free/Pro flavors

4.5 References

Please verify each reference and its details before final submission.

- Steinmacher, I., Graciotto Silva, M. A., Gerosa, M. A., and Redmiles, D. F. (2015). A systematic literature review on the barriers faced by newcomers to open source software projects. Information and Software Technology.
- Xiao, W., He, H., Xu, W., Tan, X., Dong, J., and Zhou, M. (2022). Recommending good first issues in GitHub OSS projects. International Conference on Software Engineering (ICSE).
- Lappas, T., Liu, K., and Terzi, E. (2009). Finding a team of experts in social networks. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
- Zhang, Y., and Chen, X. (2020). Explainable recommendation: A survey and new perspectives. Foundations and Trends in Information Retrieval.
- Wei, J., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. Advances in Neural Information Processing Systems (NeurIPS).